EMMANUEL BURUVURU

SOFTWARE ENGINEER

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Quatre Bornes, Mauritius

SKILLS

- Programming Languages: Java, Python, JavaScript, TypeScript
- Backend Engineering: Django, Spring Boot, Node.js, REST APIs, WebSockets, Webhooks, API Integrations, MCP servers
- Authentication & Security: JWT, Cookies,, Secure API Design, RBAC
- Frontend: React, React Native (Mobile), HTML, CSS
- Databases: PostgreSQL, MYSQL, MongoDB, Query Optimization, Data Modeling
- Cloud & DevOps: Oracle Cloud, AWS, Azure, Docker, GitHub Actions, CI/CD
- Data & AI: ETL Pipelines, Data Engineering, ML, RAG, Vectorization
- Cybersecurity Tools: Check Point, Rapid7, Microsoft Intune, Vulnerability Assessment

EDUCATION

BACHELOR OF SCIENCE INFORMATION TECHNOLOGY

Curtin University 2024

CERTIFICATION

- Cisco Certified Network certifications
- Oracle Cloud certifications

LANGUAGES

English - Fluent
French - Intermediate
German - Intermediate

AI-focused Software Engineer with experience developing full-stack applications, backend APIs, reporting automations, data pipelines, and secure AI-ready platforms. Skilled in Java, Python, Node.js, React Native, MongoDB, PostgreSQL, Oracle cloud, AWS, Azure, Docker, CI/CD, RAG, and vectorization. Experienced in building scalable systems that improve operational efficiency, data accuracy, and business continuity.

EXPERIENCE

SOFTWARE ENGINEER

Africa Clinical Research Network

Jul 2025 - Date

- Developing AI-ready operational applications processing 5K+ records/day, improving workflow efficiency by 25%+
- Managing AI-ready integrations across 10+ ERP modules, streamlining workflows, and reducing manual processing by 40%
- Designing AI-ready data pipelines and ETL workflows, improving data accuracy and processing speed by 30%+

SOFTWARE DEVELOPER / ASSOCIATE CYBERSECURITY ENGINEER

Ignite Africa

Jun 2024 - Jun 2025

- Developed reporting automations and conducted security assessments across 4 companies, remediating 50+ critical vulnerabilities before deployment.
- Built Power Apps for business continuity and reduced incidents by 30–40% across 1000+ endpoints using Check Point and Rapid7.

SOFTWARE ENGINEER

Accenture (Part Time Project)

Feb 2023 - Nov 2023

- Built full-stack application features using Spring Boot, MongoDB, Node.js, and modern frontend technologies.
- Developed and deployed secure internal learning platform modules, improving reliability, usability, and deployment workflows.



EMMANUEL BURUVURU

PERSONAL STATEMENT

I have spent the past three years building the infrastructure that machine learning depends on, and I now want to build the models themselves. As a Software Engineer at Africa Clinical Research Network (ACRN), I design and maintain the data pipelines and ETL workflows behind operational healthcare applications that process over 5,000 clinical records each day, improving data accuracy and reducing processing time by more than 30%. The work has also exposed the boundary of my current expertise: I can ensure that the data reaching a model is trustworthy, but I cannot yet determine with mathematical rigour whether the model itself is reliable, well-calibrated, and statistically sound. The MSc Applied AI and Data Science at Middlesex University is where I intend to bridge that gap.

My Bachelor of Science in Information Technology from Curtin University gave me a strong foundation in software engineering, database systems, algorithms, and software architecture, along with the discipline to design scalable, maintainable systems. Since graduating I have deliberately extended that foundation through independent study in machine learning, cloud computing, and data engineering, completing three Oracle Cloud Infrastructure certifications covering AI Foundations, Generative AI, and Data Science, alongside continuous development in Python, FastAPI, and modern machine learning workflows.

That direction is clearest in the projects I care about most. The most ambitious is AgriMesh Autonomous, an AI-driven farm intelligence platform designed for Zimbabwean agriculture, combining autonomous agents, geospatial analysis, environmental modelling, and machine learning to design farm layouts, optimise crop selection, and recommend livestock placement. Building it has shown me that the real challenge is not writing software—it is building models that stay reliable when the data is incomplete, noisy, and constantly changing. Agricultural datasets are sparse, inconsistent, and highly seasonal, requiring the integration of environmental variables, satellite imagery, and weather patterns with very limited ground-truth observation. Developing models that remain dependable under those conditions has made the value of statistical learning, uncertainty estimation, and disciplined model evaluation concrete rather than abstract.

My machine learning model for tank failure pressure prediction reinforced the same ambition. Using engineering and process data, I built predictive models to estimate the pressure thresholds associated with structural failure, comparing multiple regression techniques and evaluating their predictive performance.

The lasting lesson concerned calibration rather than accuracy:in safety-critical environments, an overconfident model is far more dangerous than one that communicates its uncertainty honestly. I could build the models, but I lacked the mathematical framework to analyse their statistical behaviour rigorously—and that gap is a large part of why I am applying.

My professional experience has consistently moved me toward this goal. Before ACRN I worked at Ignite Africa, developing reporting automation and remediating critical cybersecurity vulnerabilities across enterprise systems, and during my software engineering placement at Accenture I contributed to secure full-stack applications for internal learning platforms. Across each role I have worked at the intersection of software engineering, cloud technologies, and data infrastructure—the systems that allow machine learning to operate in production, and the vantage point from which its limits are most visible.

Middlesex attracts me because of its emphasis on combining rigorous theoretical foundations with practical application. Modules such as Applied Data Science: From Processing to NLP and LLMs, and Machine Learning and AI for Data Science, address directly the areas in which I need to grow: statistical machine learning, advanced predictive modelling, deep learning, and responsible AI. I am especially drawn to the research dissertation, where I hope to develop AgriMesh Autonomous into a rigorously evaluated framework capable of generating interpretable, reliable agricultural recommendations for resource-constrained environments. Turning that engineering challenge into reproducible research is the outcome I most want from this degree.

The word "applied" in the programme title matters to me. I have never been motivated by theory in isolation; I value it because it produces better engineering decisions. I want the mathematical depth to build models that are not only accurate but explainable, reliable, and trustworthy. My experience has taught me how to build the infrastructure that enables intelligent systems. I now want to understand the intelligence itself.

My long-term ambition is to become a Data and Machine Learning Engineer who understands why intelligent systems behave as they do, can evaluate their limitations rigorously, and can design models that remain reliable in the imperfect data environments common across African agriculture and healthcare. My software engineering experience has given me the foundation to build scalable systems; this programme will give me the scientific depth to build intelligent ones.